

Train ticket website optimization

Trains in China is an important means of transport, especially in the Chinese Lunar New Year travelers buy tickets will be very much able to buy a train ticket is every traveler's desire. Since the end of 2011 after the implementation of network tickets, buy tickets way train ticket from the mouth only to queuing at the ticket, and gradually developed into a website or via telephone booking tickets through a variety of convenient ways. Online ticket really do give to the convenience, tickets could not see the house on the computer will be able to buy tickets, the ticket window would not have so busy, saving a lot of manpower and resources. However, in 2013 Spring Festival ticket website when there is a large number of issues, login, purchase, payment and other aspects have had problems, to bring trouble ticket buyers.

On the website booking process to make the following simplification:

- 1 A query I vote for this lookup database trains remaining tickets
- 2 to generate orders, locked remaining tickets for the purchase of a ticket, according to the ticket information and ticket information generation who paid orders
- 3, completion of the payment, the ticket purchase is complete; payment fails, ticket booking recycling continues to others

Analysis of the site where the bottlenecks are:

The first site traffic problems, visit the Web site at the same time the number of excessive server was unable respective owners request, resulting in not logged in, can not query the ticket or unable to pay more. During the Spring Festival traffic to your site is very large, the site's daily visits can refer alexa records (see Resources). In order to alleviate the problem of excessive same time to visit the site to take a timeshare purchase method, equivalent to the amount of dispersion of the request, reducing at the same time the volume of traffic problems.

The second problem is the only resource that is a ticket booking people first of all to lock a ticket, if you have too many people at the same time to rob the tickets, then the system will not be able to determine that this ticket Who locked, so that the system will make a request to retry, has been unable to achieve lock busy booking tickets will cause the system to fail. In response to this ticket only resource contention, the site has taken a queuing method, that request comes first obtain a number of holding a number waiting queue, thus avoiding a direct result of the deadlock on the ticket, get a number that is very fast , while the processing speed relative to a ticket to take a number, into the queue is relatively slow, because there are locked tickets, generate orders, testing a series of operations to pay. (This process can be compared and the bank's operating procedures, to obtain a first number of queued for processing, the service window, a processor, in accordance with the FIFO queue, the order processing).

Booking sites you want to utilize a queue to meet the current ticketing needs, annex gives the queue test data: a single server test data into the queue and a server on the queue data. Which enqueueing data sampling rate is 1%, out of a queue is a queue of all the data.

Answer the following questions based on the data

- 1, evaluate existing timeshare ticket policy is reasonable, how you can optimize (requires no more than 5 timesharing)

2, according to the queue time data to estimate the queue one day be able to handle what quantity.

3, according to the site on the total amount of data access and sharing existing website strategy, simulation day's most visited website booking request volume data to calculate how many queues to meet the needs (requirements queue queuing time less than 30 minutes) .

4, according to the optimization of the time-sharing policy recalculation 3 asked questions.

Internal Reference: Data characterization (characteristics of the data, the idle time into the queue time is less than the queue time per second into the queue data added into the queue is greater than the time-consuming and also increased queue time, fitting the two time can be obtained the maximum amount of processing per second)

[1]Site access conditions can be found in:

http://alexa.chinaz.com/Show_Cache.asp?Url=12306.cn

[2]Internet ticketing passenger train Measures:

http://www.12306.cn/mormhweb/zxdt/tlxw_tdbtz11.html

[3]Some stations on the adjustment of the Internet, telephone bookings from the sale time of the announcement:http://www.12306.cn/mormhweb/zxdt/tlxw_tdbtz56.html

[4]12306 Description: <http://baike.baidu.com/view/3188461.htm>